

Interpretability of a multi-source Land Use classification machine learning model using a SHAP approach

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Introduction

Land Use



Ortho-image



Land Cover

- Built-up areas
- Unbuilt areas
- Areas with mineral materials
- Areas with other composite materials
- Bare ground
- Water surfaces
- Snow and glaciers
- Hardwood stands
- Coniferous stands
- Mixed stands
- Shrub and sub-shrub formations
- Other woody formations
- Herbaceous formations
- Other non-woody formations



Land Use

- Agriculture
- Forestry
- Extractive activities
- Fishing and aquaculture
- Other primary production
- Secondary production
- Secondary and tertiary production and residential use
- Tertiary production
- Road networks
- Rail networks
- Aerial networks
- Inland waterway and maritime transport networks
- Other transport networks
- Logistics and storage services
- Public utility networks
- Residential use
- Areas in transition
- Abandoned areas
- No use
- Unknown use

LU classification

- Land Use cannot be determined from remote sensing data only

Land Cover:
Built-up
Land Use:
Residential



BD ORTHO © IGN

Land Cover:
Built-up
Land Use:
Tertiary
production

- Land Use classification models require **multiple data sources**, that can have **imperfections**.

Context of the thesis

- **Objective:** Classification and update of Land Use databases
- **Hypothesis:** Machine learning models can predict polygon's LU out of multiple sources.

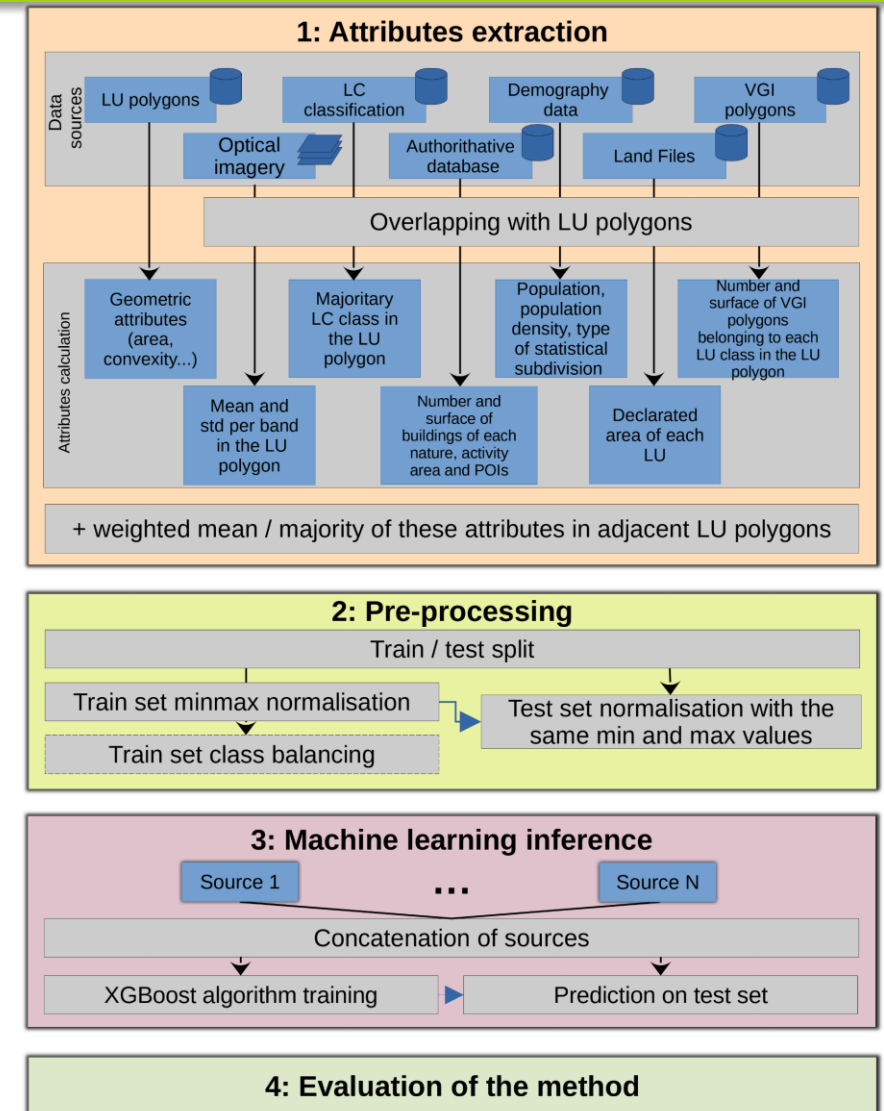
Land Use classification workflow

Land Use classification workflow (Cubaud *et al.*, 2024)

Case study: OCS GE

Base geometry: polygons

Four steps workflow:



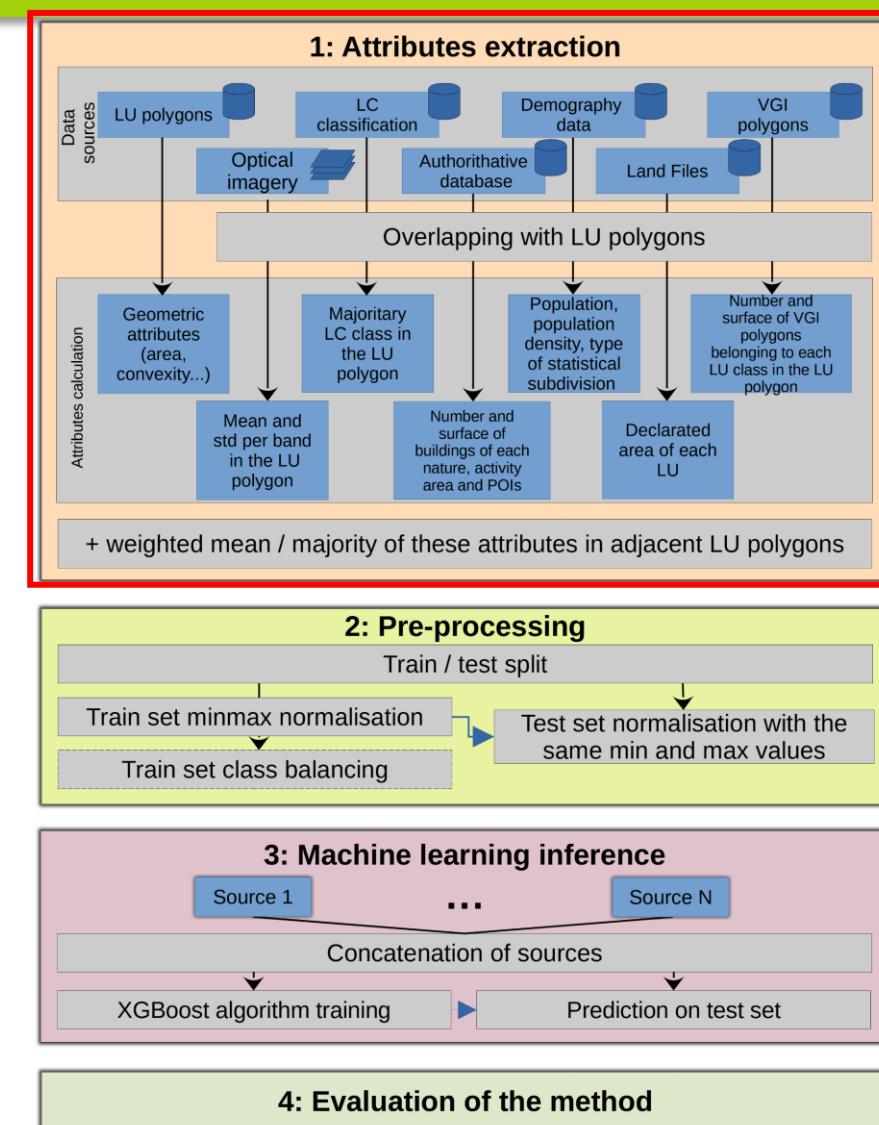
Land Use classification workflow (Cubaud *et al.*, 2024)

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Four steps workflow:

- 1) Attributes extraction from the intersections of polygons and the data sources
 - Attributes with **explicit** or **implicit** information about LU
 - **Neighboring attributes**



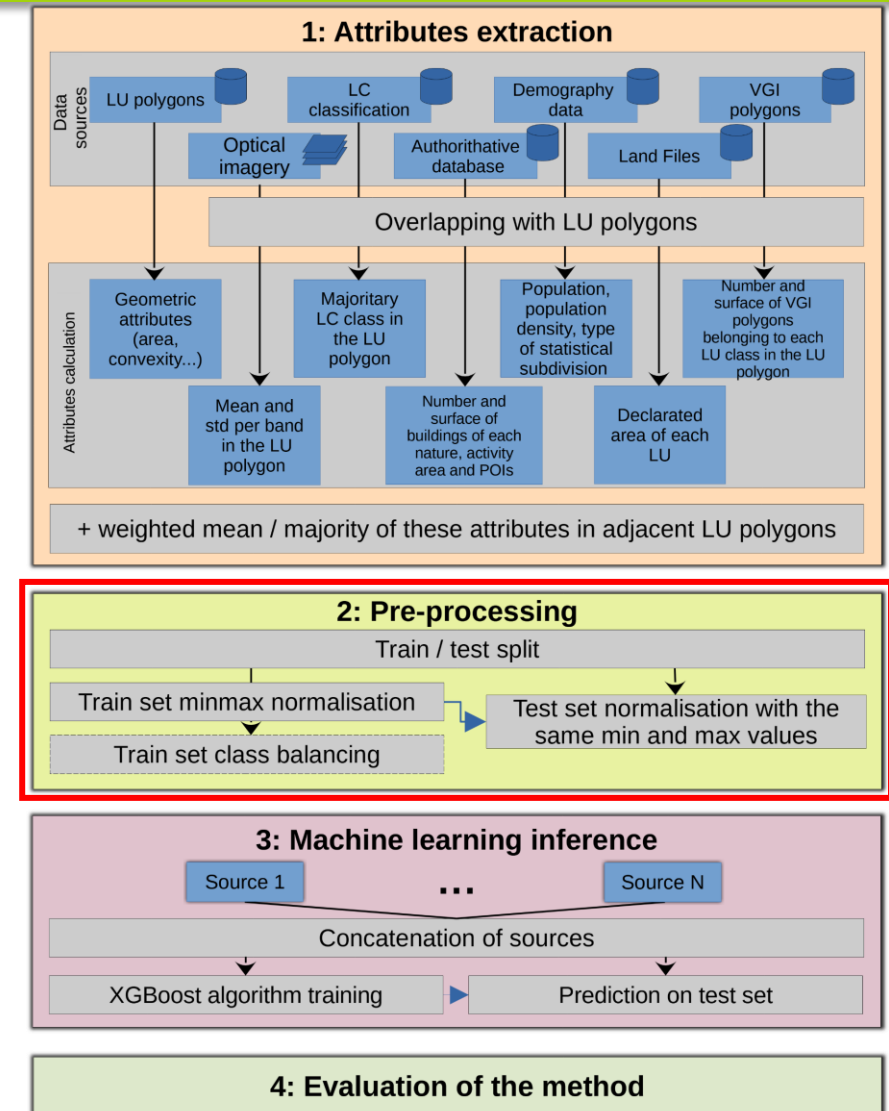
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- 2) Pre-processing e.g. class balancing



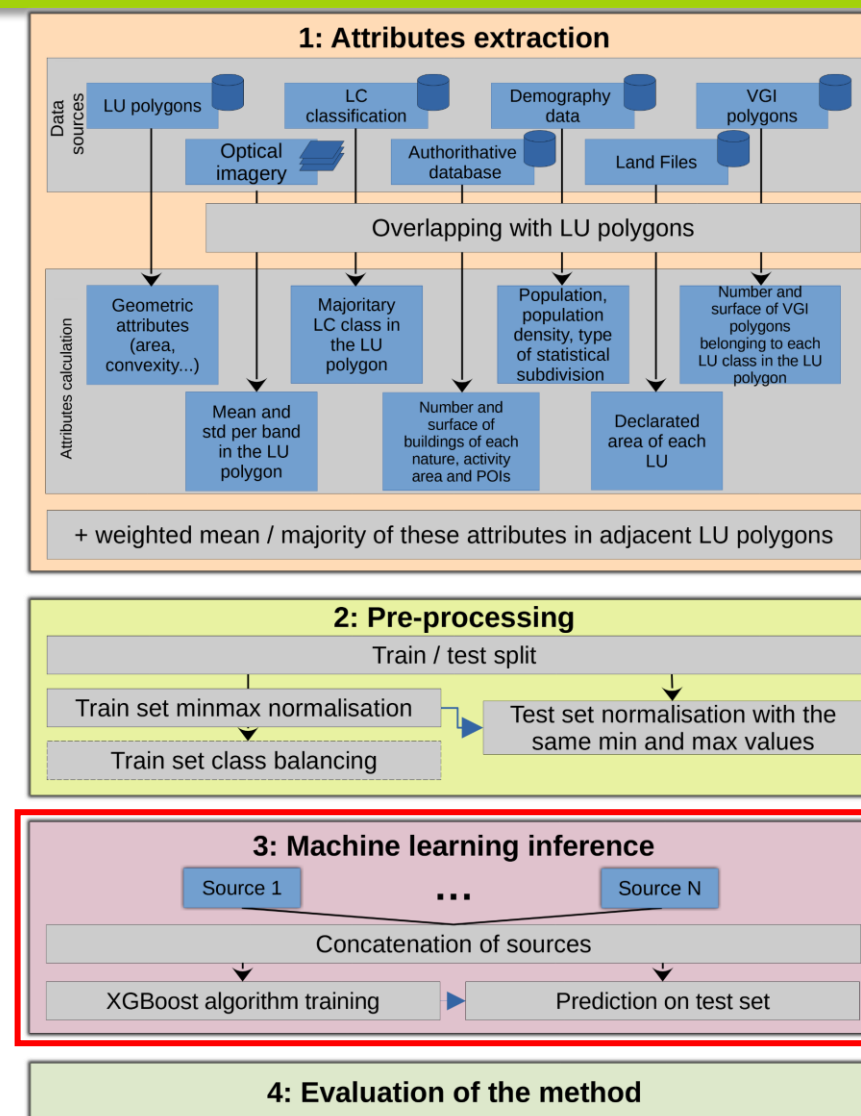
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- 3) Machine learning using **XGBoost** (Chen and Guestrin 2016)



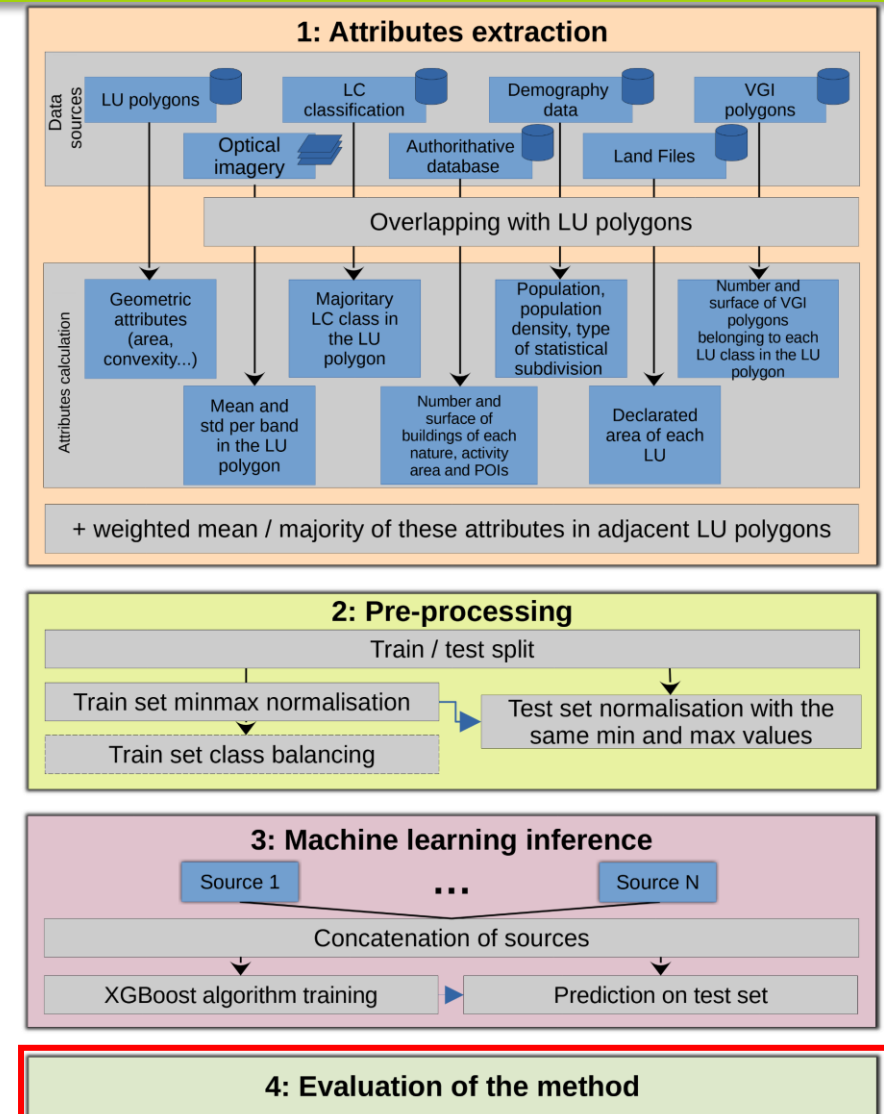
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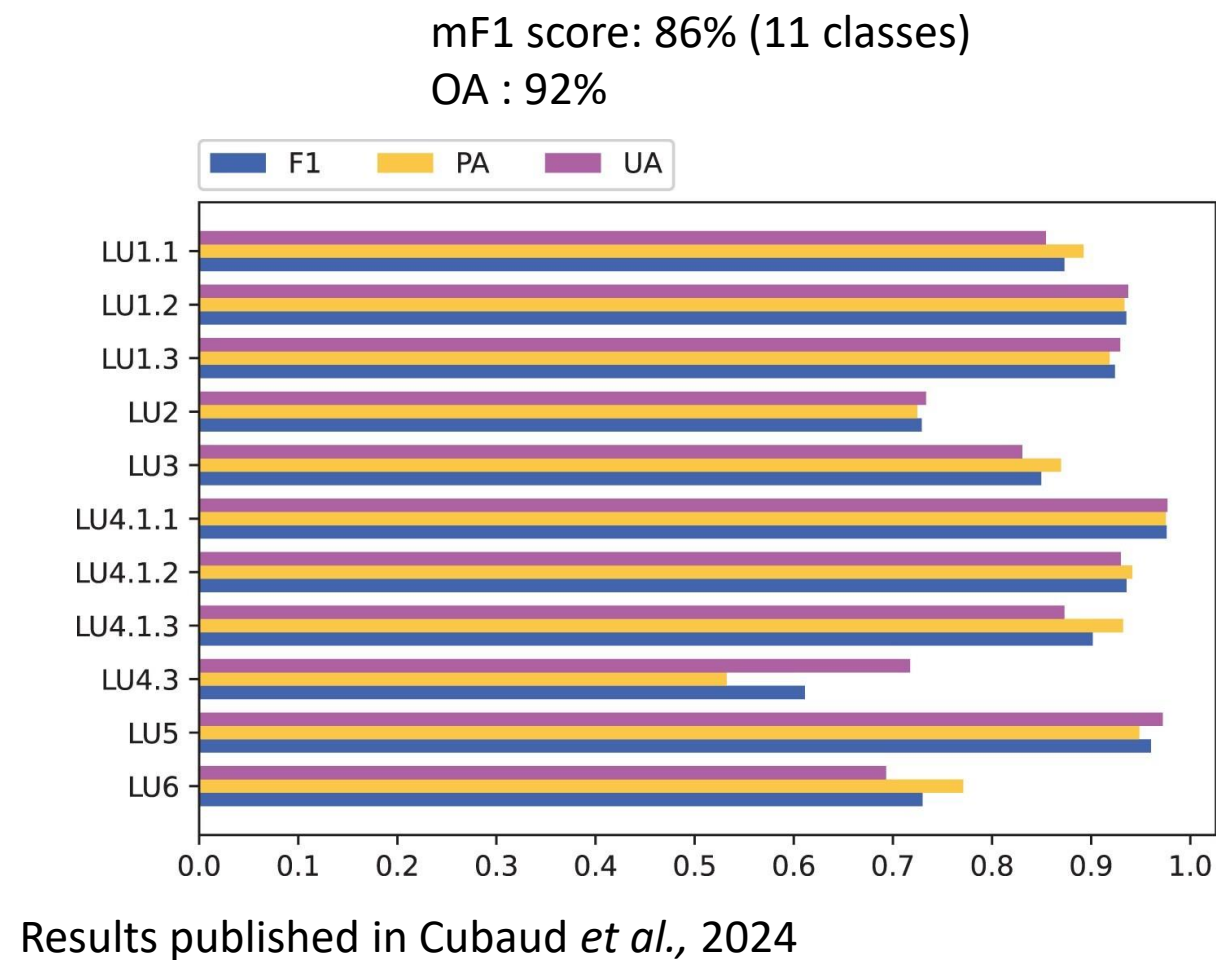
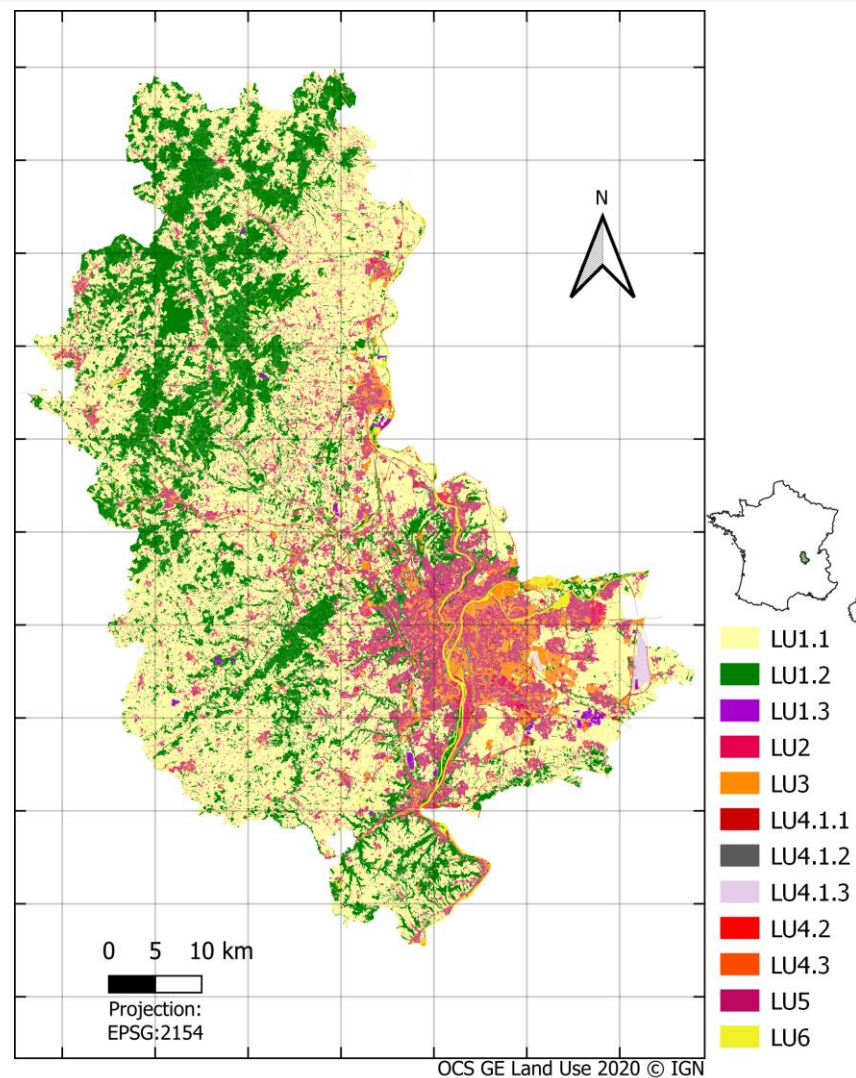
- 1) Attributes extraction from the intersections of polygons and the data sources
 - Attributes with **explicit** or **implicit** information about LU
 - **Neighboring attributes**
- 2) Pre-processing e.g. class balancing
- 3) Machine learning using **XGBoost** (Chen and Guestrin 2016)
- 4) Method Evaluation



Sources used and extracted attributes

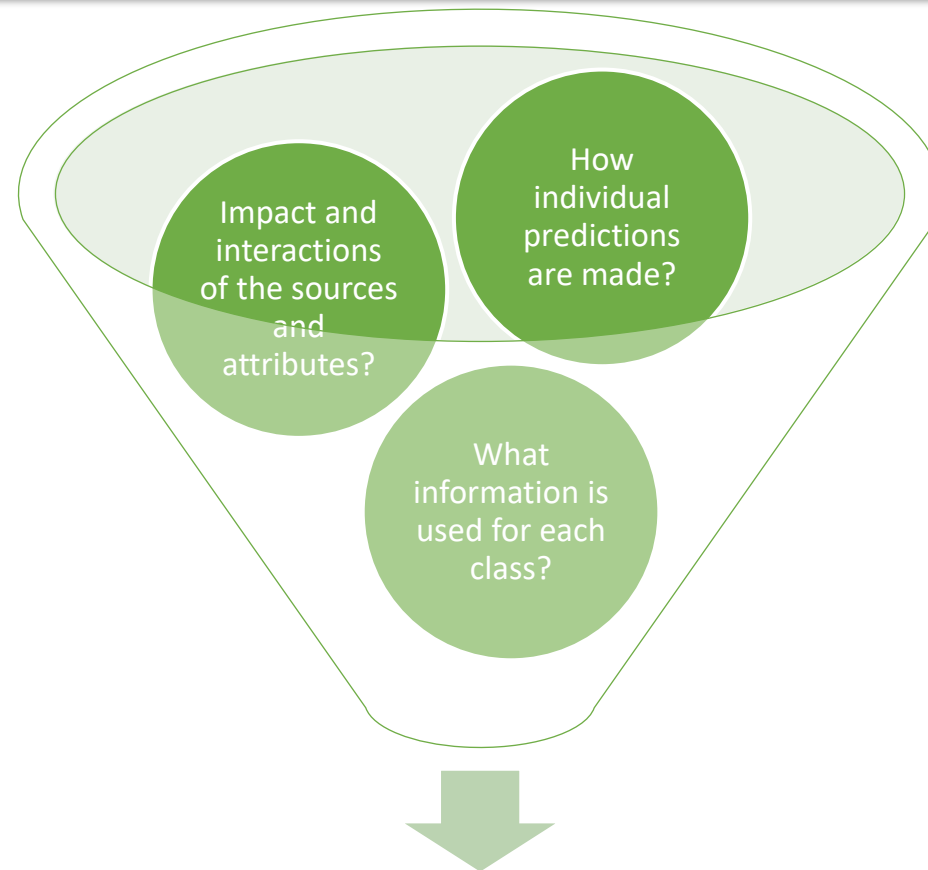
Type	Source	Number of attributes	Neighboring attributes
LU segmentation	OCS GE LU geometry	25	25
Optical imagery	BD ORTHO	8	8
LC classification	CLC	1	1
	OSO	1	1
Topographic database	BD TOPO buildings	17	17
	BD TOPO other	13	13
	RPG	1	1
	OpenStreetMap	36	36
Demographic data	INSEE	6	6
Land Files	DGFIP	18	18
Total		254	

Dataset – OCS GE LU Rhône 2020

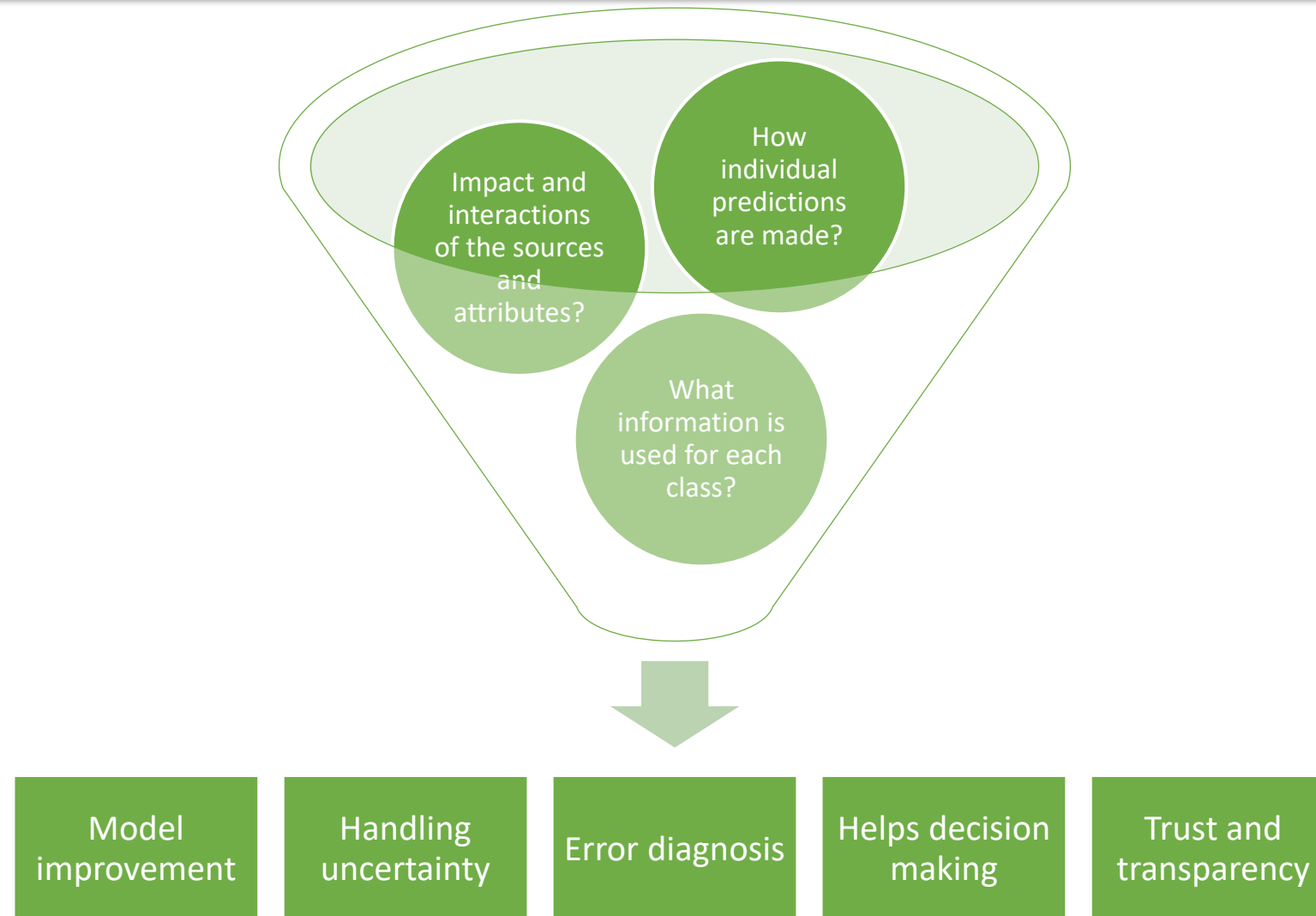


Interpretability proposed method based on SHAP

Interpretability



Interpretability

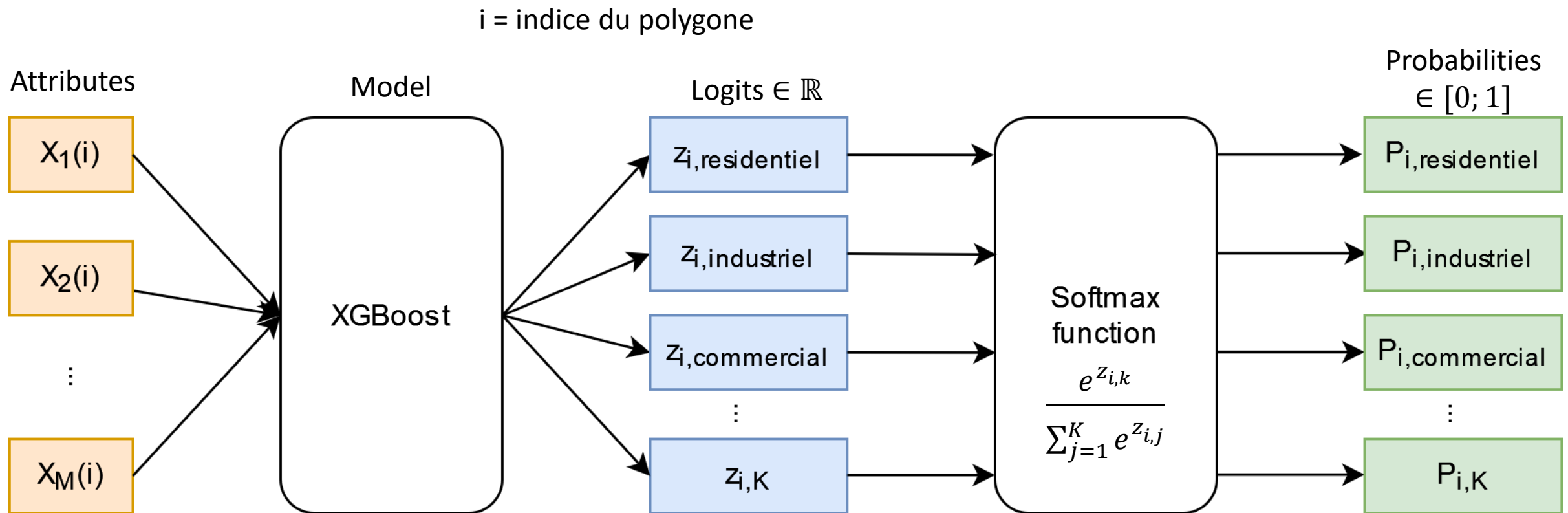


The SHAP approach

- **SHapley Additive exPlanation** values (Lundberg, Lee, 2017)
- **Individual explanations** (local) → aggregated to get **global importances** and **per class importances**
- Can be applied to **any machine learning model**
- Computed by comparing model predictions with and without each attribute, averaging over all possible attribute combinations.

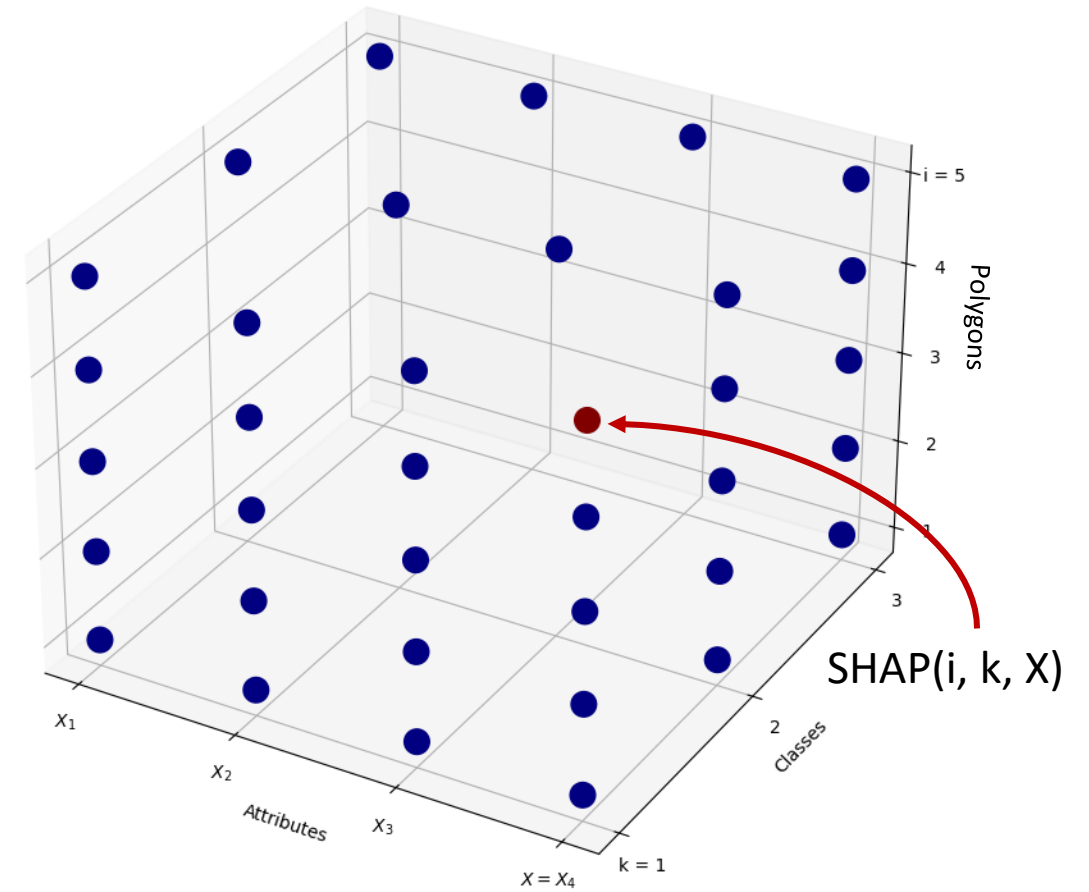
SHAP values for multiclass classification

- For multiclass classification, SHAP is based on the classes logits



SHAP values for multiclass classification

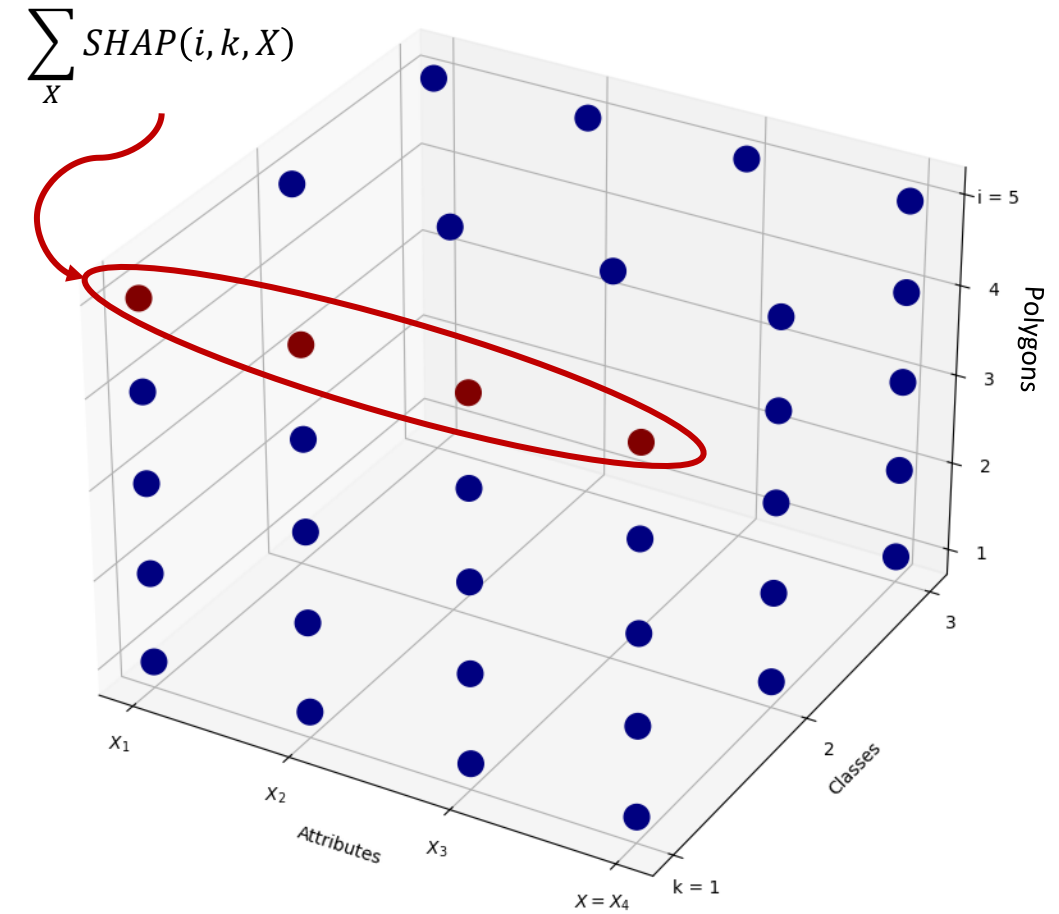
- SHAP values are defined for **each sample i** (LU polygon), for **each class k** and **each attribute X**
- SHAP values reflect the positive or negative influence of the attribute X to the likelihood of the class k for the sample i .



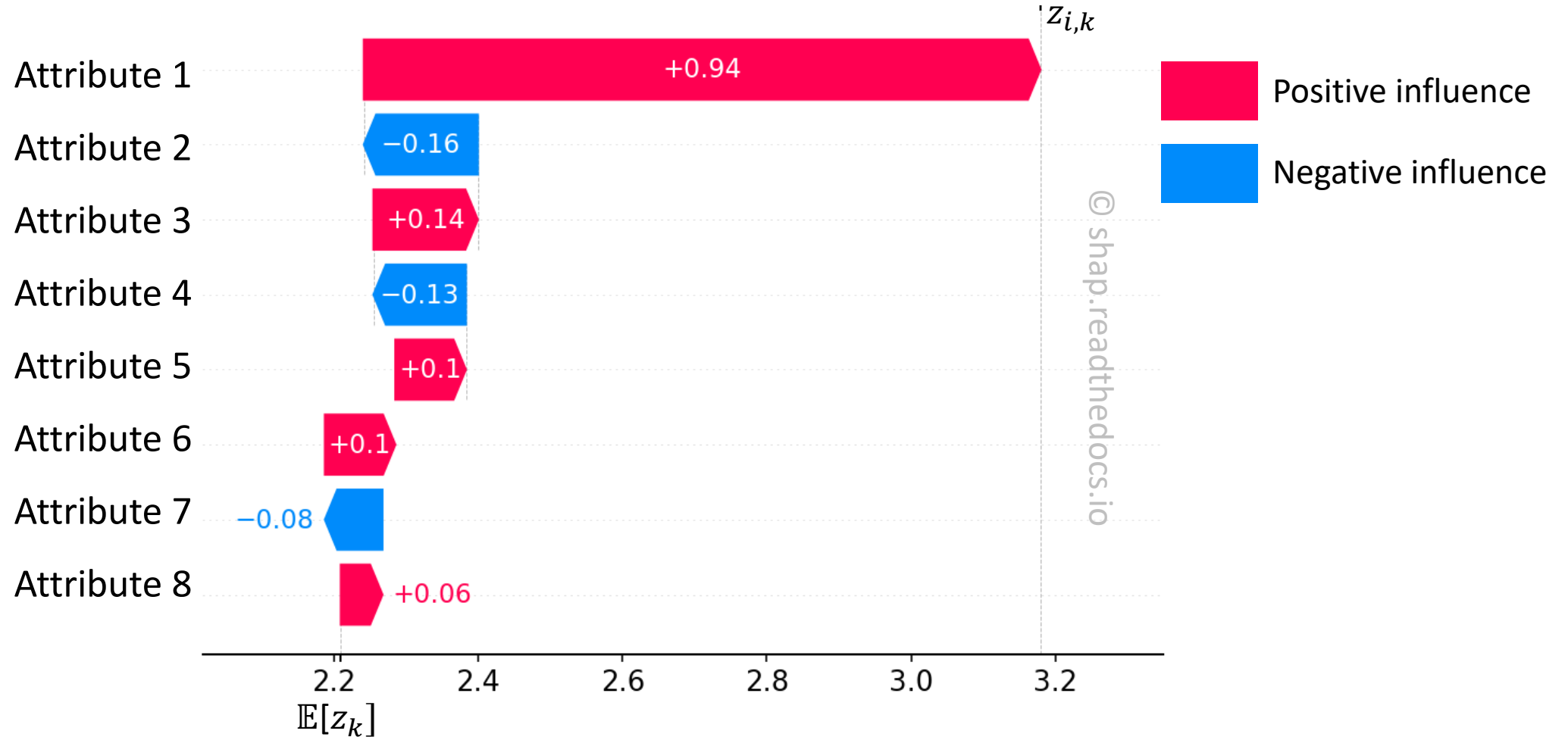
SHAP values for multiclass classification

- SHAP values are defined for **each sample i** (LU polygon), for **each class k** and **each attribute X**
- SHAP values reflect the positive or negative influence of the attribute X to the likelihood of the class k for the sample i.
- SHAP values are **additive**, which means that:

$$z_{i,k} = \mathbb{E}[z_k] + \sum_X SHAP(i, k, X)$$



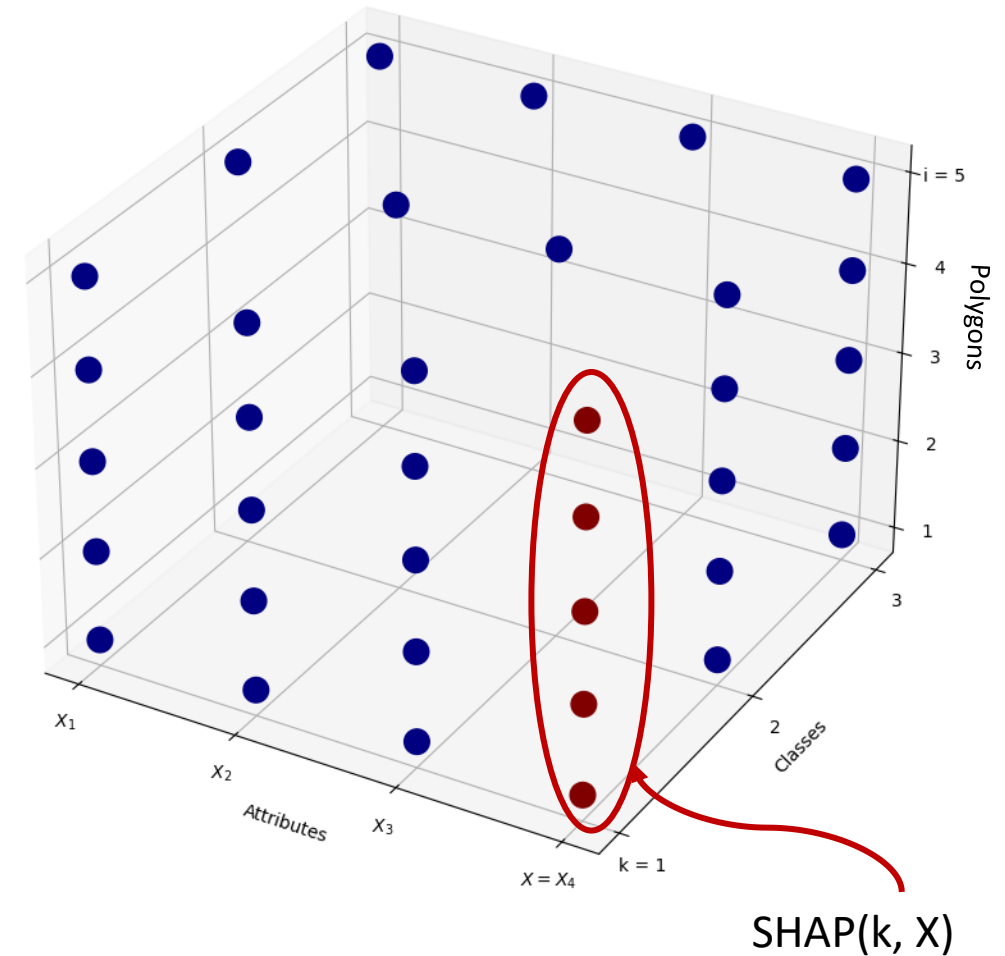
SHAP values for multiclass classification



SHAP values for multiclass classification

- The importance of an attribute X for a class k can be computed as :

$$SHAP(k, X) = \frac{1}{N} \sum_{i=1}^N |SHAP(i, k, X)|$$



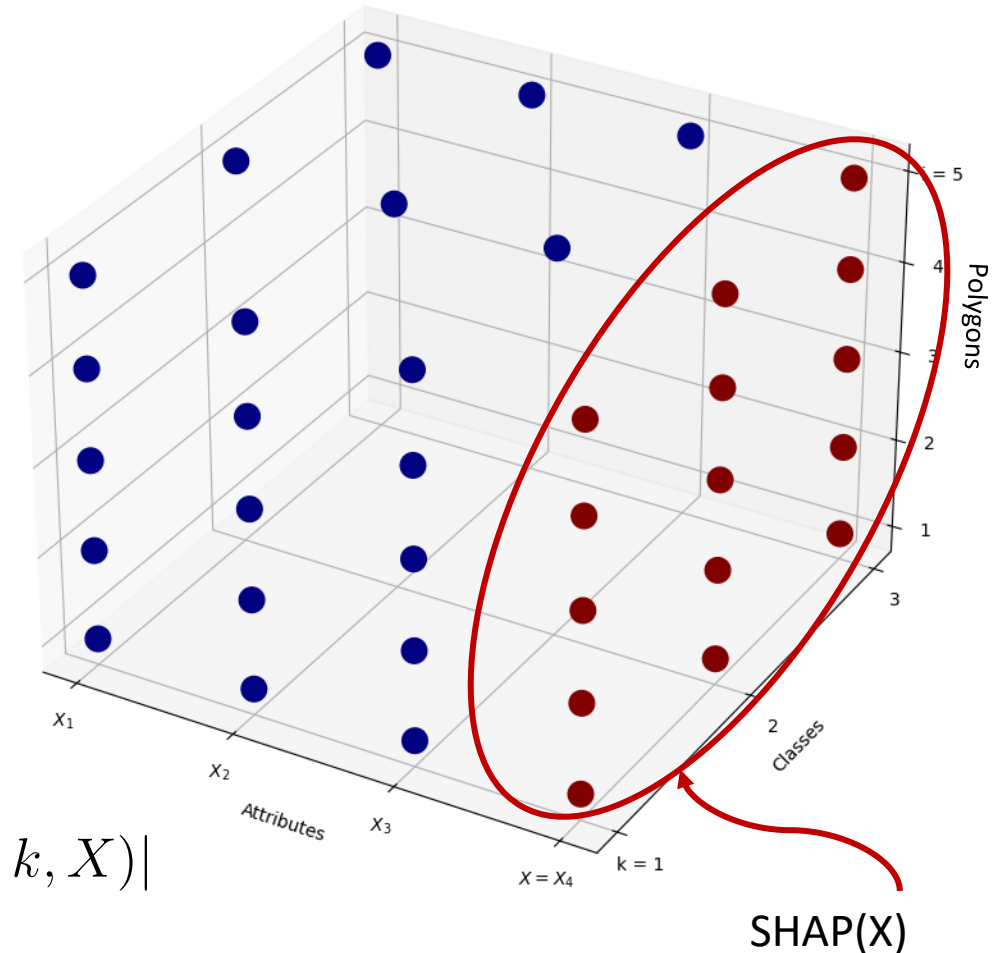
SHAP values for multiclass classification

- The importance of an attribute X for a class k can be computed as :

$$SHAP(k, X) = \frac{1}{N} \sum_{i=1}^N |SHAP(i, k, X)|$$

- The global importance of an attribute X can be computed as :

$$SHAP(X) = \sum_{k=1}^K SHAP(k, X) = \frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K |SHAP(i, k, X)|$$



Interpretation method

Formalize hypothesis on how attributes affect the predictions

Compute SHAP values to verify the hypothesis

Analyze examples of accurate and misclassifications

Interpretation method

Formalize hypothesis on how attributes affect the predictions

- E.g. explicit attributes primarily contributes to their associated LU class

Compute SHAP values to verify the hypothesis

Analyze examples of accurate and misclassifications

Interpretation method

Formalize hypothesis on how attributes affect the predictions

Compute SHAP values to verify the hypothesis

- By aggregating the values per class and globally and ranking the attributes

Analyze examples of accurate and misclassifications

Interpretation method

Formalize hypothesis on how attributes affect the predictions

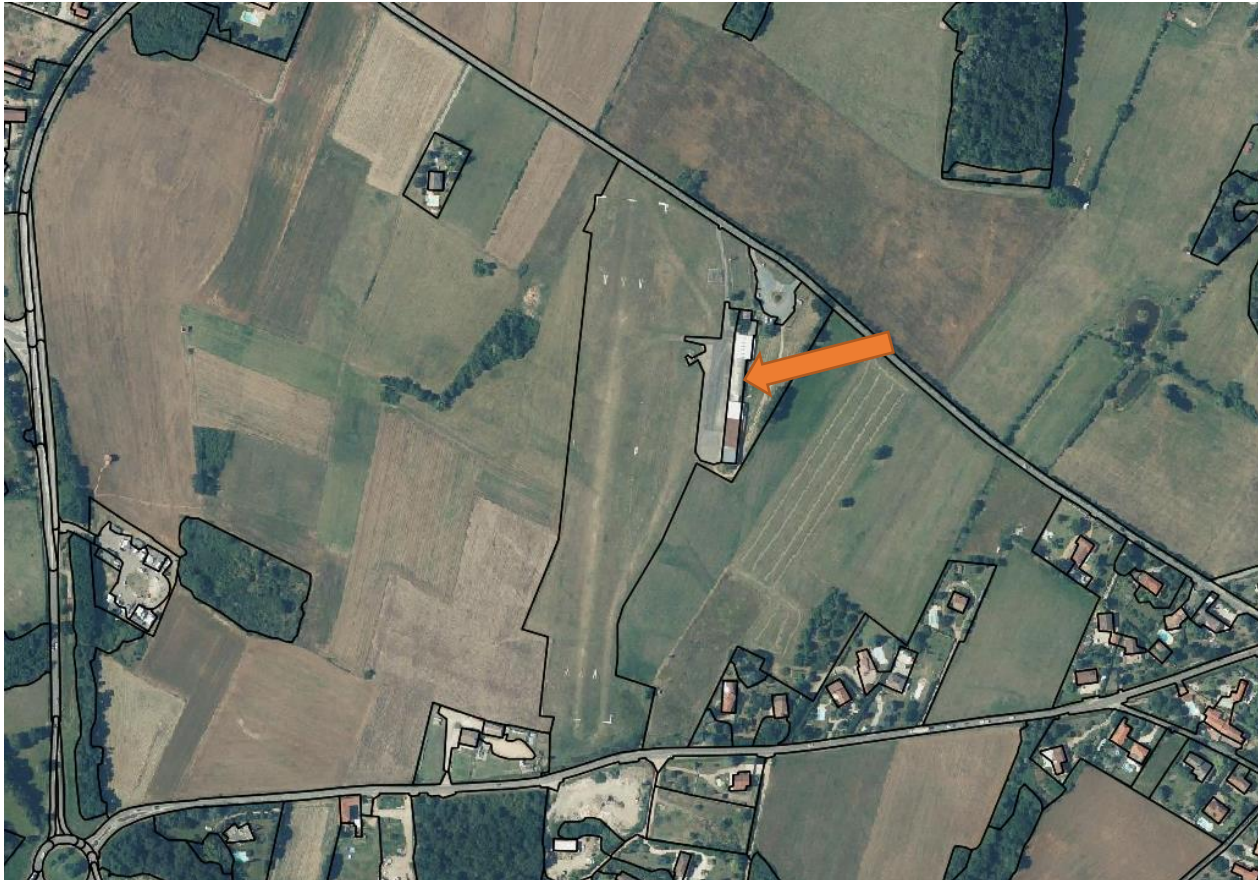
Compute SHAP values to verify the hypothesis

Analyze examples of accurate and misclassifications

- To understand why the model makes errors

Results

Example of a misclassification



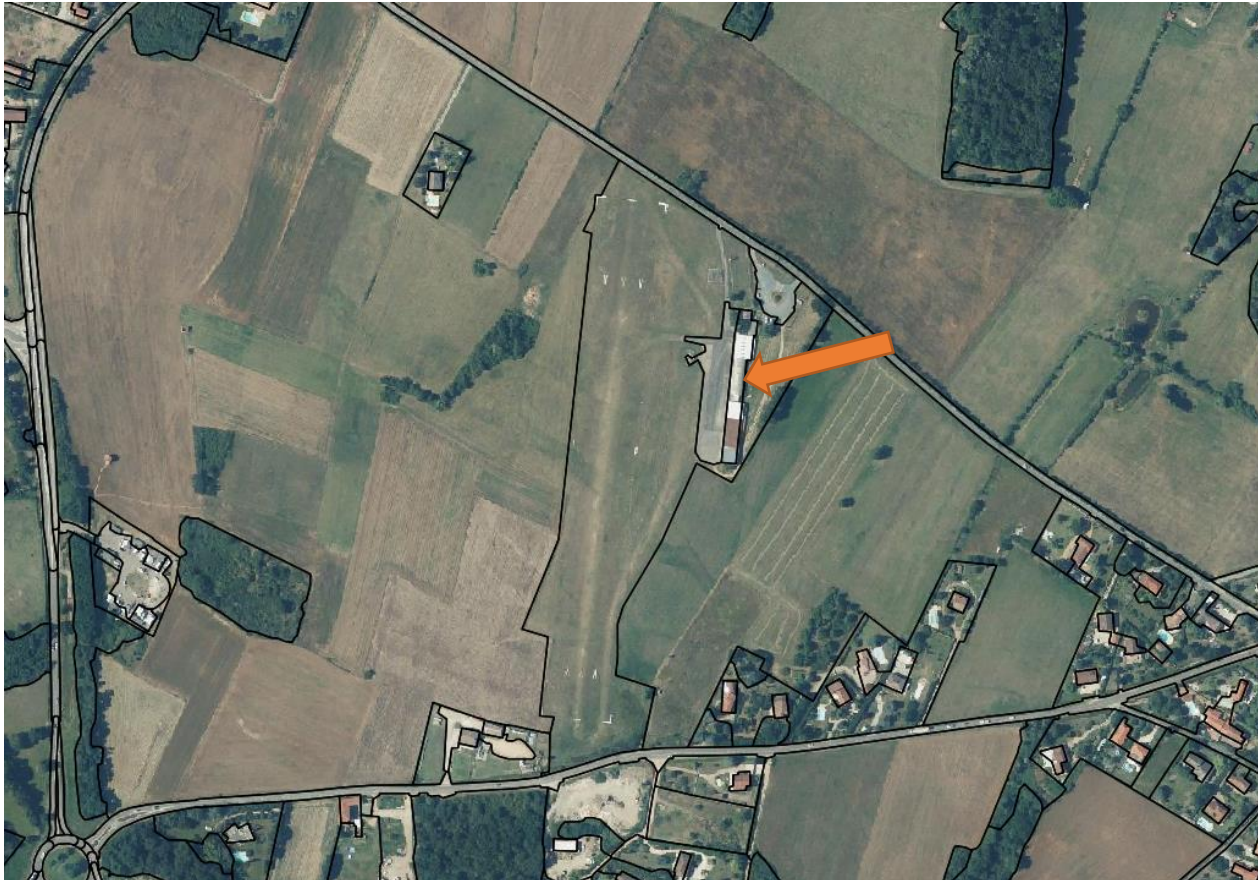
BD Ortho, OCSGE © IGN

Aerodrome of Lyon – Brindas



Predicted LU3 (tertiary production) instead of LU4.1.3
(air transport)

Example of a misclassification



BD Ortho, OCSGE © IGN

Aerodrome of Lyon – Brindas



Predicted LU3 (tertiary production) instead of LU4.1.3
(air transport)

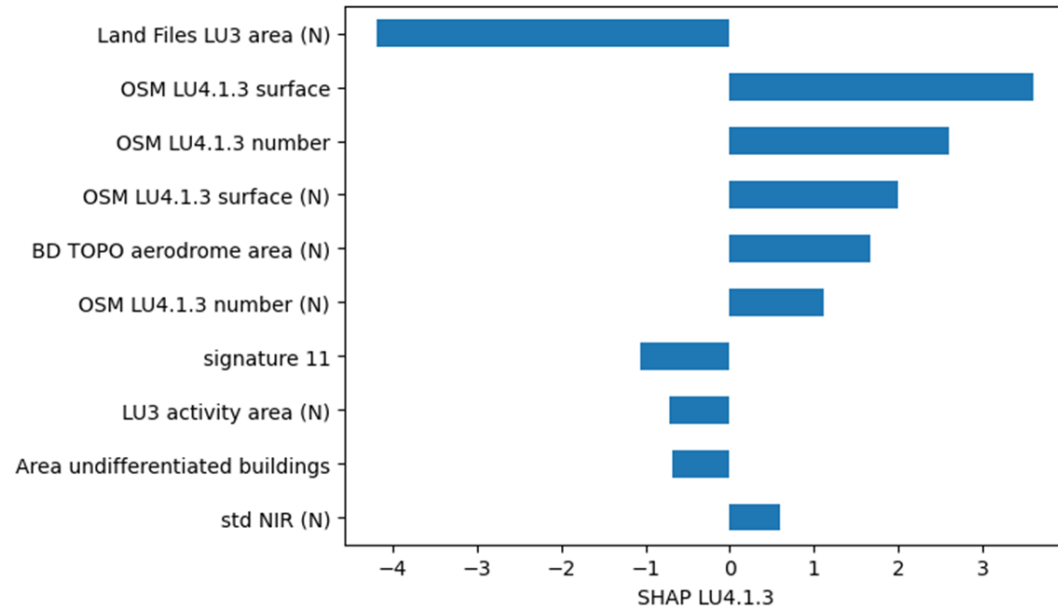
Information given by the sources:

- Aerodrome in BD TOPO and OSM
- Undifferentiated buildings in BD TOPO and one commercial building
- In land files the area is considered as tertiary

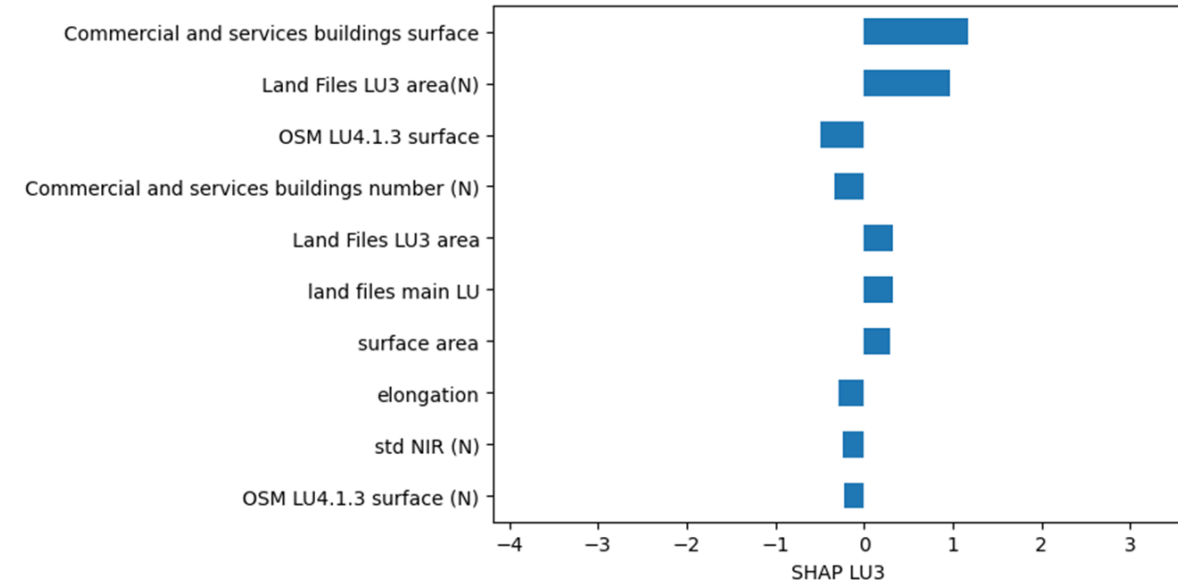
Example of a misclassification

Polygon Aerodrome of Lyon – Brindas

True class : Air transport (LU4.1.3)



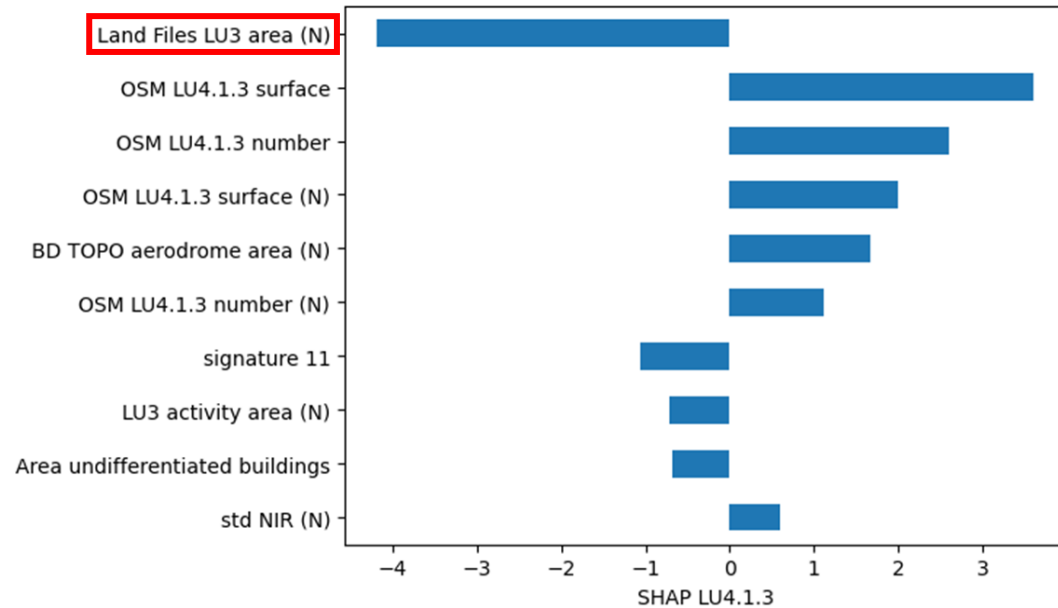
Badly predicted class : Tertiary production (LU3)



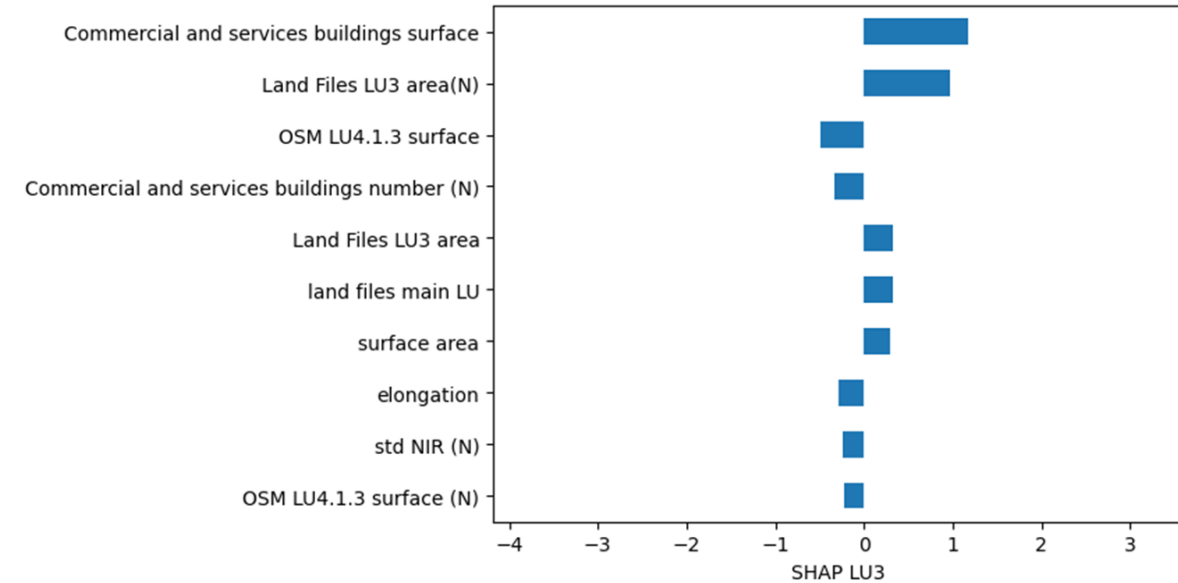
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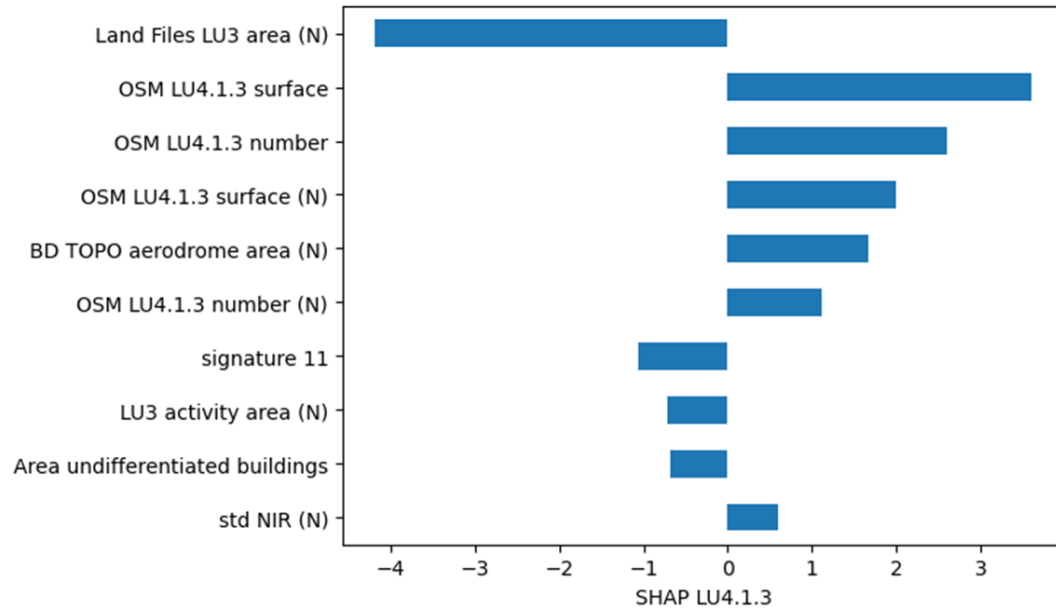
Badly predicted class : Tertiary production (LU3)



Example of a misclassification

Polygon Aerodrome of Lyon – Brindas

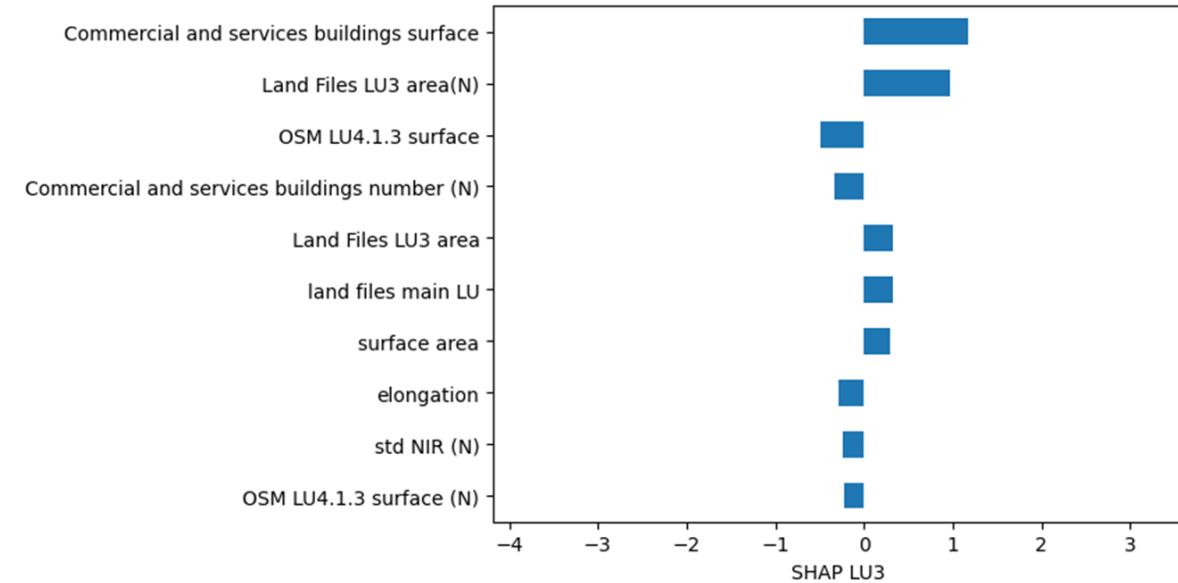
True class : Air transport (LU4.1.3)



Base probability LU4.1.3 = 0.08%
Base logits LU4.1.3 = -4.71

$\log(p) + cst$

Badly predicted class : Tertiary production (LU3)

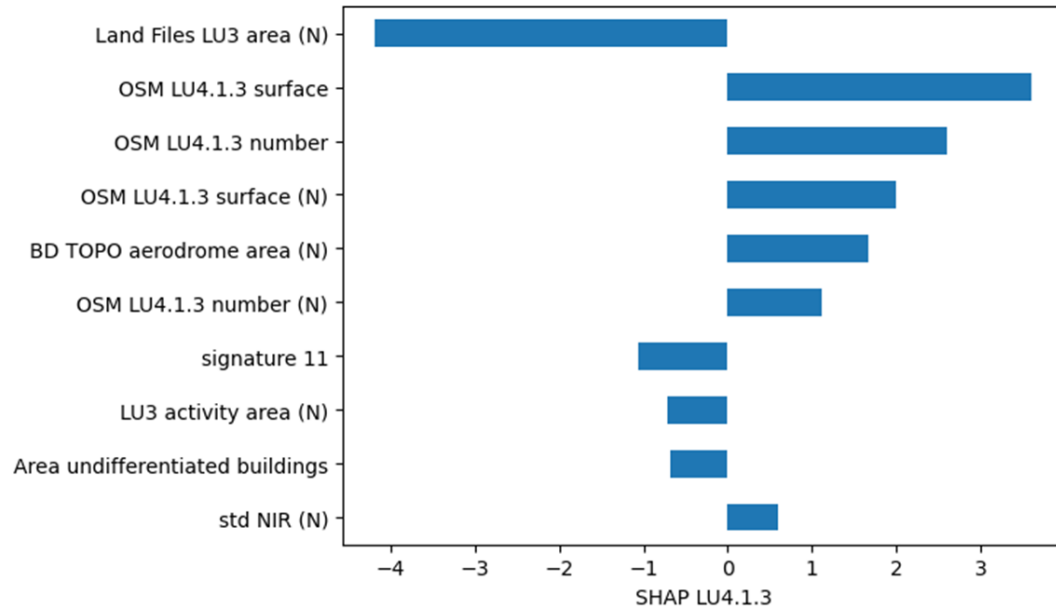


Base probability LU3 = 13.1%
Base logits LU3 = 0.36

Example of a misclassification

Polygon Aerodrome of Lyon – Brindas

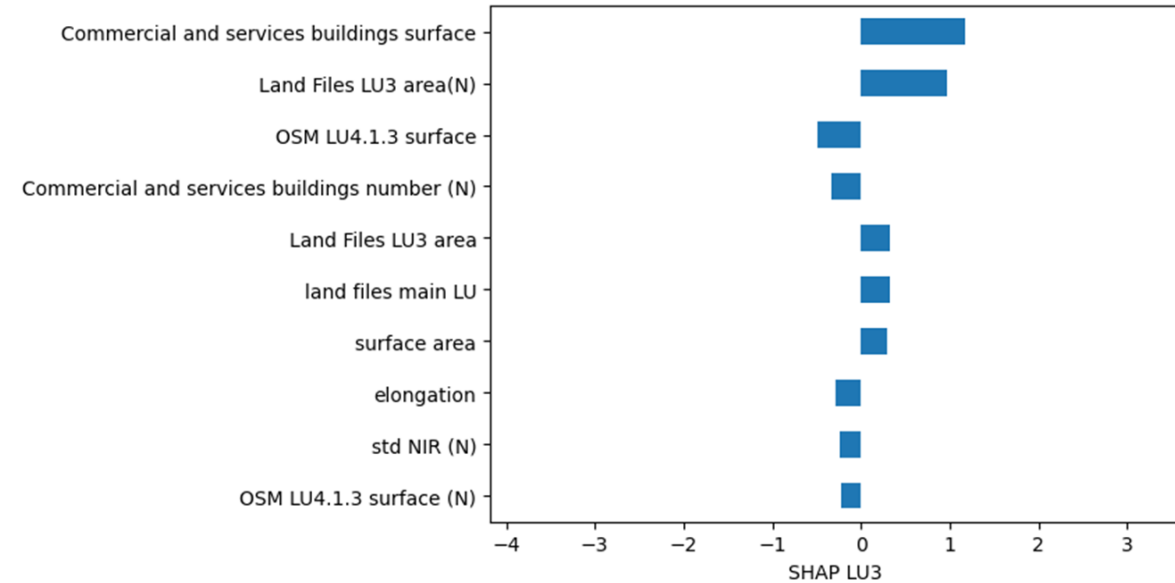
True class : Air transport (LU4.1.3)



Base probability LU4.1.3 = 0.08%
 Base logits LU4.1.3 = -4.71
 $\sum_X \text{SHAP}(\text{aerodrome, LU4.1.3, } X) = 2.19$

$\Rightarrow \log(p) + \text{cst}$

Badly predicted class : Tertiary production (LU3)

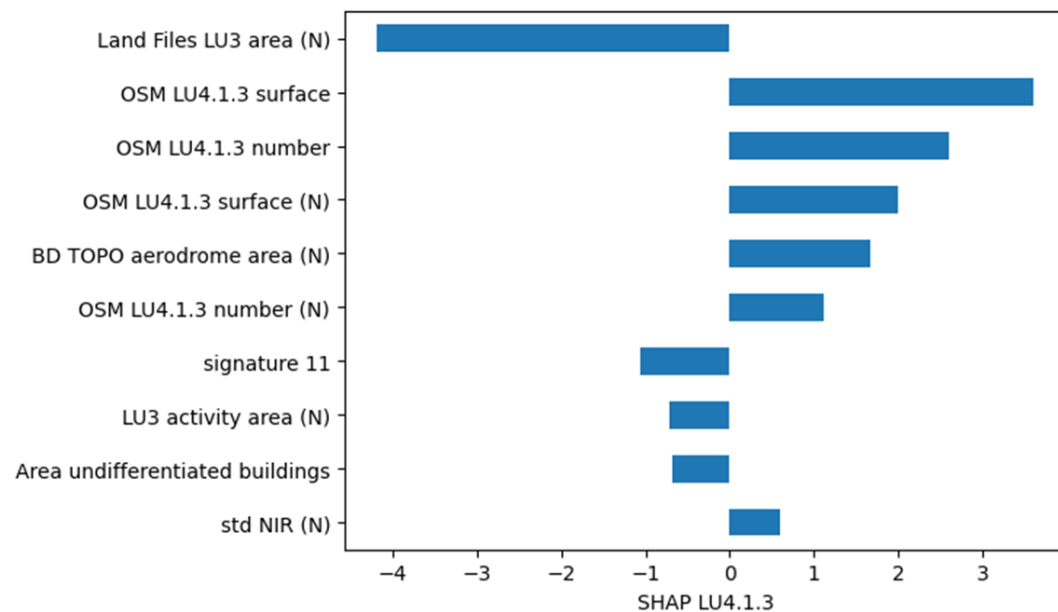


Base probability LU3 = 13.1%
 Base logits LU3 = 0.36
 $\sum_X \text{SHAP}(\text{aerodrome, LU3, } X) = 1.35$

Example of a misclassification

Polygon Aerodrome of Lyon – Brindas

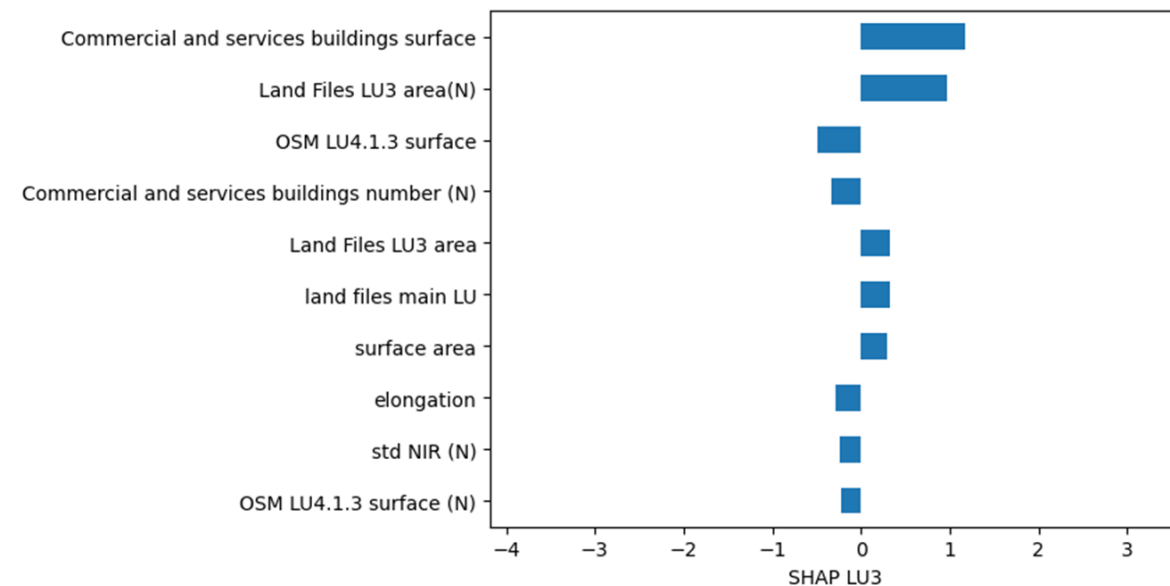
True class : Air transport (LU4.1.3)



Base probability LU4.1.3 = 0.08%
 Base logits LU4.1.3 = -4.71
 $\sum_X \text{SHAP}(\text{aerodrome, LU4.1.3, } X) = 2.19$
 Final logits = -2.59
 Probability LU4.1.3 = 1.31%

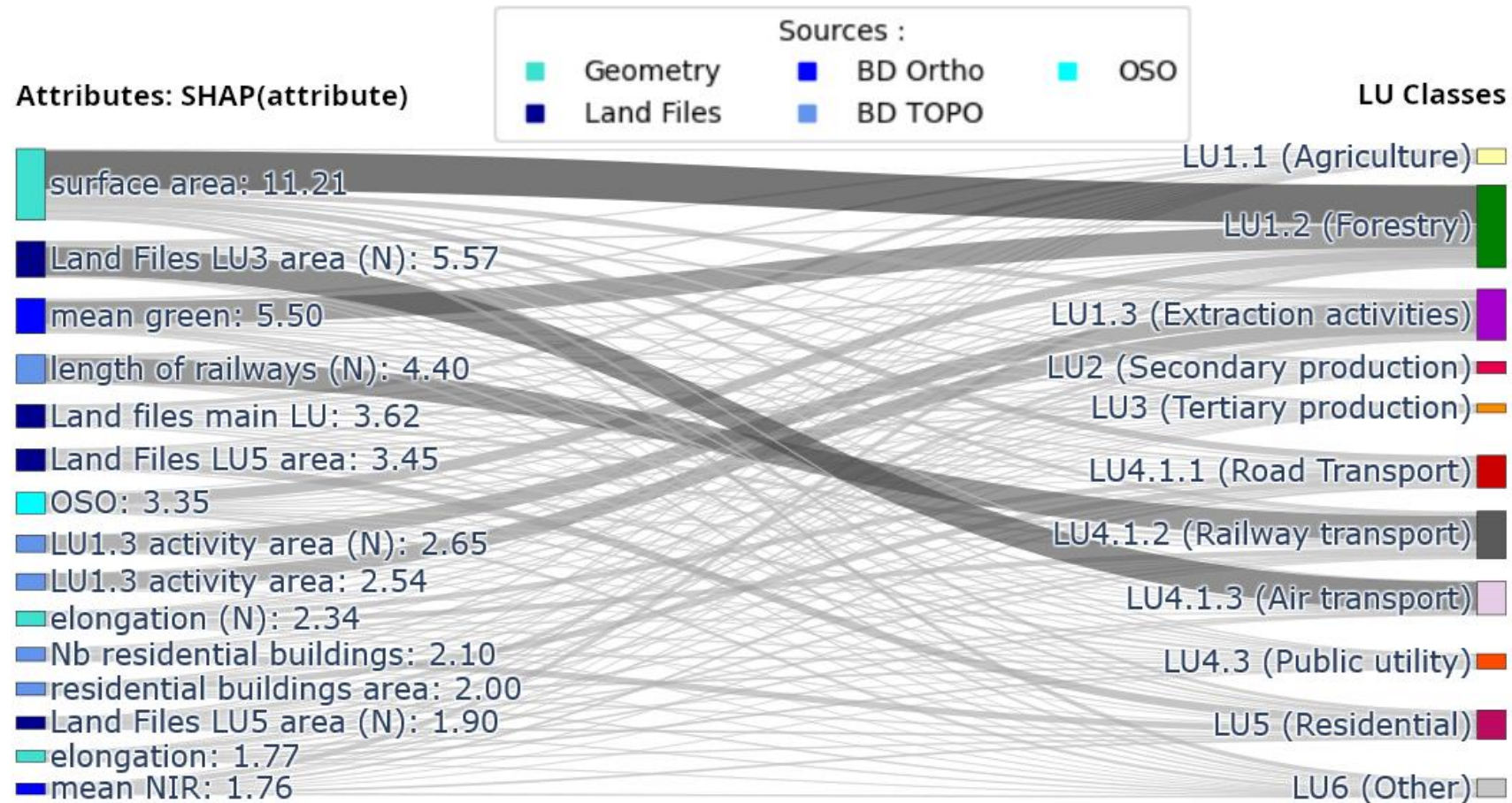
$\log(p) + \text{cst}$
 softmax

Badly predicted class : Tertiary production (LU3)

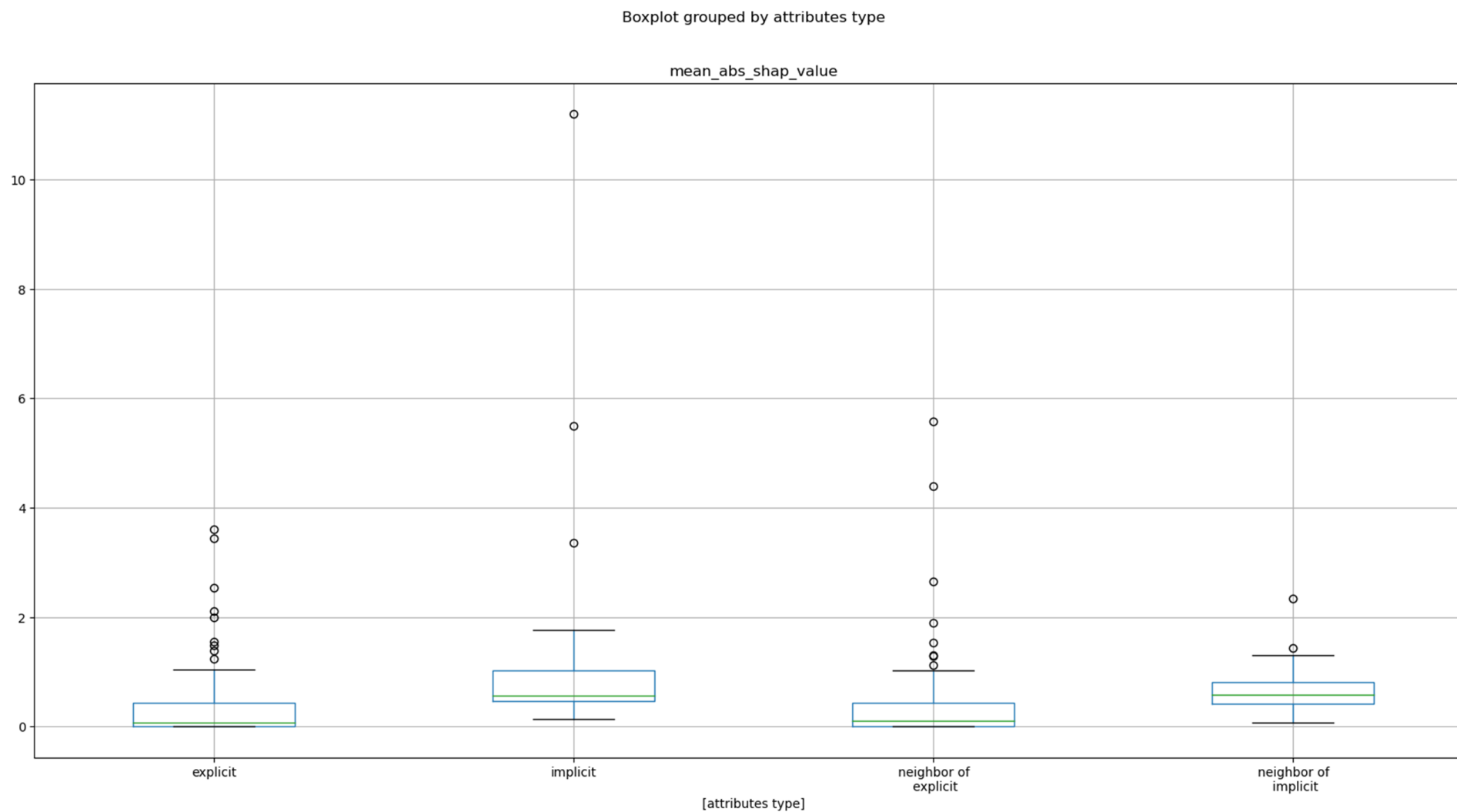


Base probability LU3 = 13.1%
 Base logits LU3 = 0.36
 $\sum_X \text{SHAP}(\text{aerodrome, LU3, } X) = 1.35$
 Final logits = 1.71
 Probability LU3 = 91%

Link between attributes and classes



Implicit VS explicit attributes



Conclusion

Conclusion

Method to **better understand LU classification models**

SHAP approach is a **powerful tool** that provides relevant insights at both **local** and **global** scale

Extension to other tasks : **transferability, change detection**

Model the derived knowledge in a machine-readable format **to improve the reuse** of the LU classification workflow

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Lundberg S. M., Lee S.-I. (2017). A unified approach to interpreting model predictions. In I. Guyon et al. (Eds.), Advances in neural information processing systems 30 , p. 4765–4774. Curran Associates, Inc.

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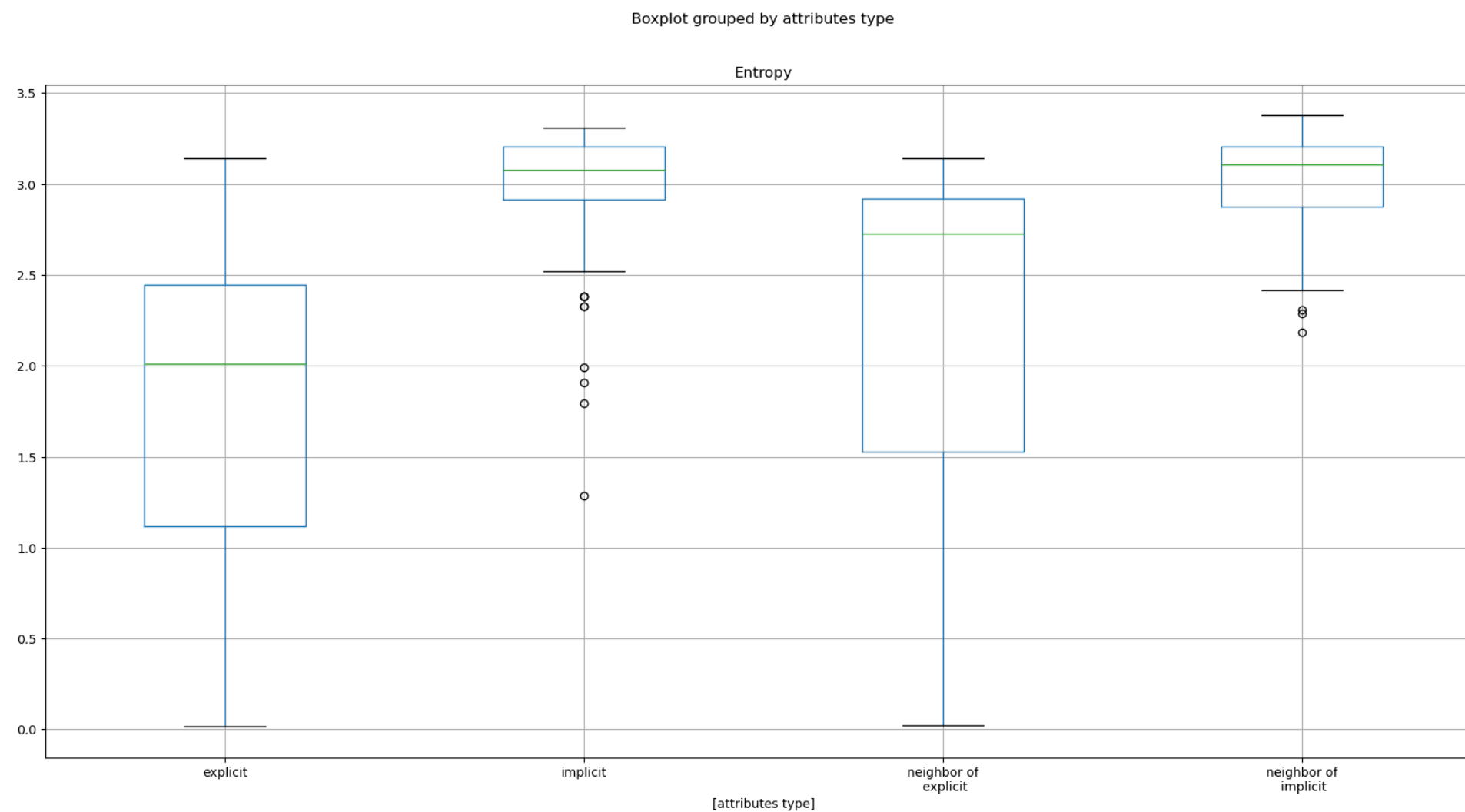
Liu L., Olteanu-Raimond A.-M., Jolivet L., Le Bris A., See L. (2021). A data fusion-based framework to integrate multi-source VGI in an authoritative land use database. , vol. 14, no 4, p. 480–509

Ribeiro M. T., Singh S., Guestrin C. (2016). "Why Should I Trust You?": Explaining the Predictions of Any Classifier. In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, p. 1135–1144. Association for Computing Machinery.

Tu Y., Chen B., Zhang T., Xu B. (2020). Regional Mapping of Essential Urban Land Use Categories in China: A Segmentation-Based Approach. , vol. 12, no 7, p. 1058.

Thank for your attention !

Implicit VS explicit attributes



Per class importance

Class	LU1.1 (Agriculture)	LU1.2 (Forestry)	LU1.3 (Extraction activities)	LU2 (Secondary production)	LU3 (Tertiary production)	LU4.1.1 (Road transport)	LU4.1.2 (Railway transport)	LU4.1.3 (Air transport)	LU4.3 (Public utilities)	LU5 (Residential)	LU6 (other)
Most important attribute	Land Files main LU	Surface Area	LU1.3 activity areas (N)	OSM area of LU2 landuse polygons	Land Files main LU	Surface Area	Length of railways (N)	Land Files LU3 area (N)	Building height (N)	Land Files LU5 area	Surface Area

Per class importance

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- The most important attribute for **more than half** of the classes is an **explicit attribute**.

Per class importance

Class	LU1.1 (Agriculture)	LU1.2 (Forestry)	LU1.3 (Extraction activities)	LU2 (Secondary production)	LU3 (Tertiary production)	LU4.1.1 (Road transport)	LU4.1.2 (Railway transport)	LU4.1.3 (Air transport)	LU4.3 (Public utilities)	LU5 (Residential)	LU6 (other)
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- The most important attribute for more than half of the classes is an explicit attribute.
- The most important attribute for 4 out of 11 classes is a **neighboring attribute**.

Per class importance

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- The most important attribute for more than half of the classes is an explicit attribute.
- The most important attribute for 4 out of 11 classes is a neighboring attribute.
- 66% of the attributes with **explicit information for one class** have their highest SHAP value **for this class**. The others have their highest SHAP value for another class for which the model has **confusions** with this class.

SHAP values for multiclass classification

- SHAP values are defined for **each sample** i (LU polygon), for **each class** k and **each attribute** X
- SHAP values reflect the positive or negative influence of the attribute X to the likelihood of the class k for the sample i .
- SHAP values are **additive**, which means that:

$$z_{i,k} = \mathbb{E}[z_k] + \sum_X SHAP(i, k, X)$$

SHAP - formule

$$SHAP(i, k, X) = \sum_{S \subseteq F \setminus \{X\}} \omega(S) \mathbb{E}[z_k | X' = x'_i \forall X' \in S \cup \{X\}] - \mathbb{E}[z_k | X = x'_i \forall X' \in S]$$

SHAP - formule

$$SHAP(i, k, X) = \sum_{S \subseteq F \setminus \{X\}} \omega(S) \mathbb{E}[z_k | X' = x'_i \forall X' \in S \cup \{X\}] - \mathbb{E}[z_k | X = x'_i \forall X' \in S]$$

Somme sur tous les sous-ensembles S d'attributs qui ne contiennent pas X

SHAP - formule

Poids qui dépend de la taille de S

$$SHAP(i, k, X) = \sum_{S \subseteq F \setminus \{X\}} \omega(S) \mathbb{E}[z_k | X' = x'_i \forall X' \in S \cup \{X\}] - \mathbb{E}[z_k | X = x'_i \forall X' \in S]$$

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Poids qui dépend de la taille de S

Somme sur tous les sous-ensembles S d'attributs qui ne contiennent pas X

Espérance du logit de la classe k lorsque les valeurs des attributs X' appartenant à S, et l'attribut X sont les mêmes que pour le polygone i (moyenne sur les valeurs prises par les autres attributs)

SHAP - formule

$$SHAP(i, k, X) = \sum_{S \subseteq F \setminus \{X\}} \omega(S) \mathbb{E}[z_k | X' = x'_i \forall X' \in S \cup \{X\}] - \mathbb{E}[z_k | X = x'_i \forall X' \in S]$$

Poids qui dépend de la taille de S

Somme sur tous les sous-ensembles S d'attributs qui ne contiennent pas X

Espérance du logit de la classe k lorsque les valeurs des attributs X' appartenant à S, et l'attribut X sont les mêmes que pour le polygone i (moyenne sur les valeurs prises par les autres attributs)

Espérance du logit de la classe k lorsque les valeurs des attributs X' appartenant à S sont les mêmes que pour le polygone i (moyenne sur les valeurs prises par les autres attributs, dont l'attribut X)